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Project-Based Learning

Phase-I Evaluation

FOOD FLOW: Intelligent Food Recommendation and Route Optimization System

Team ID: DSCPP-III-2026-T165

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

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Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T165
Semester: 3rd
Section:
Domain: Data Structures & C++

Team Members

1. Ipsita Bansal – 2510013430
2. Surya Pratap Tyagi – 2510012313
3. Shubhda Joshi – 2510012509
4. Sehar Gupta – 2510350019

Mentor

Dr.Pawan Kumar Mishra **Interactions:** _____

Objectives

Project Objectives

1. Current limitation / gap.
2. Impact on users or organizations.
3. Limitations of existing approaches.
4. Compare food places based on rating, distance, preparation time and user choice.
5. Provide dine-in, takeaway/drive-through and delivery options.
6. Add login, ordering, feedback and order history features.

Key Objective: To develop a working C++ console prototype for food selection and route planning.

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Proposed Solution

Proposed Approach

1. **User Input:** Gets the user's food choice, location, and service type.
2. **Food Processing:** Looks for available food items and restaurant details.
3. **Recommendation:** Picks and compares food options based on what the user wants.
4. **Route Planning:** Uses a map and Dijkstra's method to find the best short route.
5. **Final Output:** Shows the selected food place, service option and route details.

Problem–Solution Mapping

Problem	Solution
1) Too many choices	Filter and compare options
2) Nearest place may not suit	Use rating, distance and preference
3) Route planning needed	Graph + Dijkstra
4) Different service choices	Dine-in, takeaway and delivery
5) User records needed	Login, feedback and history

Main Idea: Feed back is collected Help the user select an appropriate restaurant and obtain the necessary route information.

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Integration of PBL Course Concepts

Course	Concepts Applied	Project Use
Data Structures with C	Graphs, Linked Lists, Stacks, Queues, Searching, Sorting and Hashing	Data storage, searching and route optimization
OOPs with C++	Classes, Objects, Functions and STL Containers	User, restaurant and service modules

Data Structures Role

- Graph stores locations and routes.
- Dijkstra's algorithm finds the shortest path.
- Searching and sorting help process food places.
- Linked lists, queues and stacks support data handling.
- Hash maps can provide quick record lookup.

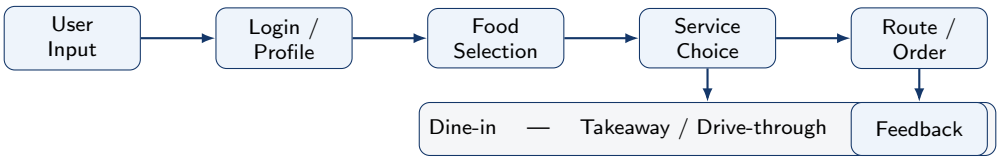
OOPs Role

- Classes represent users, restaurants and orders.
- Objects store and work with related data.
- Functions divide the system into manageable modules.
- STL containers support collection and processing of data.

Integration: C++ classes and functions use the selected data structures and algorithms to implement the recommendation, service and route modules in one system.

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System Architecture / Workflow



Major Components

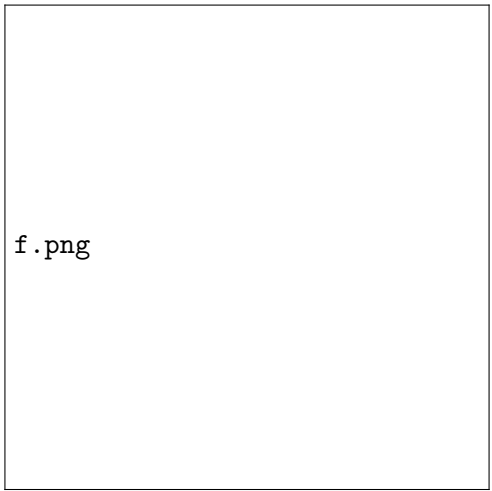
- Login and Password
- Food Selection and Recommendation
- Dine-in
- Takeaway / Drive-through
- Delivery
- Route Optimization
- Feedback

System Workflow

- User logs in and selects food preferences.
- Food options are filtered and processed.
- User selects the required service.
- Order and route information is processed.
- Feedback is collected after the service or order.

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Flow diagram



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Technology Stack & Methodology

Technology Stack

- **Language: C++** : chosen for building the system with an object-oriented approach
- **Data Structures: Graph, Vector, Queue, Stack, Linked List, Hash Map**: support how routes are represented and how data is stored and handled.
- **Algorithm: Dijkstra's Algorithm**: applied to determine the shortest possible route
- **IDE: Visual Studio Code**: used to write and test the code
- **Interface: Console-based**: provides a straightforward way for users to interact with the system.

Development Methodology

1. **Requirement Analysis**: Understand what users need and define the project's core features.
2. **System Design**: Lay out the workflow, structure the modules, and decide on the data structures to use.
3. **Module Development**: Build each module separately — login, food, service, route, and feedback.
4. **Module Integration**: Bring the individual modules together into a single, functioning system.
5. **Testing and Debugging**: Check each module on its own, then test the full user journey end to end.

Reason for Selection: The technology and data structures fit the project's needs and bring together concepts from both PBL subjects.

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Initial Progress & Team Contribution

Progress Achieved

- Problem and motivation identified
- Project requirements discussed
- System workflow planned
- Core data structures selected
- Dijkstra's algorithm selected for route optimization
- Main modules divided among team members
- Initial module development started

Individual Contribution

Member	Role	Contribution
Ipsita	Team Lead	Login password Feedback module
Shubhda	Member	Dine-in module
Sehar	Member	Takeaway and Drive-through module
Surya	Member	Delivery module

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Roadmap, Expected Outcomes & References

Roadmap

1. Complete individual modules
2. Integrate all modules
3. Test recommendation and route logic
4. Test complete user workflow
5. Finalize project and documentation

Expected Outcomes

- Working C++ console prototype
- Food recommendation based on project factors
- Shortest-route calculation
- Takeaway, dine-in and delivery options
- Login, feedback and order history

References

1. B. Stroustrup, *The C++ Programming Language*.
2. Cormen et al., *Introduction to Algorithms*.
3. C++ Standard Library Documentation.
4. E. W. Dijkstra, "A Note on Two Problems Connected with Graphs," 1959.

Thank You
Questions & Suggestions